

DEPARTMENT OF

AUTOMOBILE

ENGINEERING

VISION OF THE DEPARTMENT

To be a global center of excellence in automotive engineering through world-class education and interdisciplinary research

MISSION OF THE DEPARTMENT

- To provide quality education with the highest learning standards for the sustainable design and development of world-class automobiles.
- To strengthen research, develop innovative applications of automotive technology, and promote entrepreneurship by instilling ethical leadership qualities.

**B. TECH.
(AUTOMOBILE ENGINEERING)**

B. TECH. (AE)

PROGRAM EDUCATIONAL OBJECTIVES

PEO-I: Provide students with a strong foundation in mathematical, scientific, and engineering fundamentals, enabling them to formulate, analyze, and solve engineering problems.

PEO-II: Enable students to apply, integrate, and synthesize knowledge of automotive technology to analyze and solve multi-disciplinary engineering challenges.

PEO-III: Equip graduates with strong teamwork and collaboration skills for a successful career in the automotive industry, enabling them to contribute effectively to national and global environments.

PEO-IV: Conduct research and development using a multi-disciplinary framework, aiming to resolve automotive engineering issues of societal relevance with sustainable solutions.

PEO-V: Practice professional, ethical, and social responsibilities for a successful engineering career.

B. TECH. (AE)

PROGRAM OUTCOMES

PO-1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO-2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4).

PO-3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5).

PO-4: Conduct Investigations of Complex Problems: Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO-5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including

prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6).

PO-6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7)

PO-7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9).

PO-8: Individual and Collaborative Team work: Function effectively as an individual, and as a member or leader in diverse/multi-disciplinary teams.

PO-9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences.

PO-10: Project Management and Finance: Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.

PO-11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii)

adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8).

B. TECH. (AE)

PROGRAM SPECIFIC OUTCOMES

PSO-1: Apply fundamental automobile engineering knowledge, along with modeling, design, and simulation tools, to effectively evaluate and analyze automotive systems.

PSO-2: Address multi-disciplinary problems in the field of automotive engineering by conducting innovative and sustainable research.

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY HYDERABAD

B. TECH. I YEAR

AUTOMOBILE ENGINEERING

I SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT101	Matrices and Calculus	3	1	0	4	4
25BS1PH101	Applied Physics	3	0	0	3	3
25ES1DS101	Problem Solving Through C	3	0	0	3	3
25PC1AE101	Manufacturing Technology	3	0	0	3	3
25HS1EN101	English for Skill Enhancement	3	0	0	3	3
25BS2PH101	Applied Physics Laboratory	0	0	2	2	1
25ES2DS101	Problem Solving Through C Laboratory	0	0	2	2	1
25HS2EN101	English Language and Communication Skills Laboratory	0	0	2	2	1
25ES2ME101	Engineering Workshop	0	0	2	2	1
25MN6HS101	Induction Programme	2	0	0	2	0
Total		17	1	8	26	20

II SEMESTER

R25

Course Code	Title of the Course	L	T	P/D	CH	C
25BS1MT105	Ordinary Differential Equations and Vector Calculus	2	1	0	3	3
25BS1CH102	Engineering Chemistry	3	0	0	3	3
25ES1DS102	Data Structures through C	3	0	0	3	3
25ES1EE104	Elements of Electrical and Electronics Engineering	2	0	0	2	2
25ES1AE101	Fundamentals of Engineering Mechanics	3	0	0	3	3
25ES3ME102	Engineering Graphics	0	0	6	6	3
25BS2CH101	Engineering Chemistry Laboratory	0	0	2	2	1
25ES2DS102	Data Structures through C Laboratory	0	0	2	2	1
25ES2EE104	Elements of Electrical and Electronics Engineering Laboratory	0	0	2	2	1
25MN6HS102	Environmental Science	2	0	0	2	0
Total		15	1	12	28	20

L – Lecture T – Tutorial P – Practical D – Drawing
 C – Credits SE – Sessional Examination CA – Class Assessment
 SEE – Semester End Examination D-D – Day to Day Evaluation
 CP – Course Project PE – Practical Examination

CH – Contact Hours/Week
 ELA – Experiential Learning Assessment
 LR – Lab Record

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1MT101) MATRICES AND CALCULUS

TEACHING SCHEME		
L	T/P	C
3	1	4

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn the concept of a rank of the matrix and applying this concept to check the consistency of a system of linear equations
- To learn the concept of eigen values and eigen vectors and to reduce the quadratic form to canonical form
- To know the geometrical approach to the mean value theorems and their application to the mathematical problems
- To find the maxima and minima of function of two and three variables
- To evaluate multiple integrals and their applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Compute the rank of a matrix and analyze the solution of a system of linear equations

CO-2: Calculate the Eigenvalues and Eigen vectors and Reduce the quadratic form to canonical form using orthogonal transformations

CO-3: Solve the applications on the mean value theorems

CO-4: Find the extreme values of functions of two and three variables with/ without constraints

CO-5: Evaluate the multiple integrals and apply the concept to find areas, volumes

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	2	1	1	1	1	-	2	1	2	2	1
CO-2	3	2	2	1	2	1	1	-	2	1	2	2	1
CO-3	3	2	2	1	1	1	1	-	2	1	2	2	1
CO-4	3	2	2	1	2	1	1	-	2	1	2	2	1
CO-5	3	2	2	1	2	1	1	-	2	1	2	2	1

UNIT-I:

Matrices: Rank of a matrix by Echelon form and Normal form – Inverse of Non-singular matrices by Gauss-Jordan method. System of linear equations: Solving system of Homogeneous and Non-Homogeneous equations, Gauss Seidel Iteration Methods.

UNIT-II:

Eigenvalues and Eigenvectors: Eigenvalues – Eigenvectors and their properties – Diagonalization of a matrix – Cayley-Hamilton Theorem (without proof) – finding inverse and power of a matrix by Cayley-Hamilton Theorem – Quadratic forms and

Nature of the Quadratic Forms – Reduction of Quadratic form to canonical forms by Orthogonal Transformation.

UNIT-III:

Single Variable Calculus: Limits and Continuous function and its properties. Mean value theorems: Rolle's theorem – Lagrange's Mean value theorem with their Geometrical Interpretation and applications – Cauchy's Mean value Theorem – Taylor's Series (All the theorems without proof).

Curve Tracing: Curve tracing in cartesian and polar coordinates

UNIT-IV:

Partial Differentiation and Its Applications: Definitions of Limit and continuity – Partial Differentiation: Euler's Theorem – Total derivative – Jacobian – Functional dependence & independence. Applications: Maxima and minima of functions of two variables and three variables using method of Lagrange multipliers.

UNIT-V:

Multiple Integrals: Evaluation of Double Integrals (Cartesian and polar coordinates) – change of order of integration (only Cartesian form) – Change of variables for double integrals (Cartesian to polar). Evaluation of Triple Integrals – Change of variables for triple integrals (Cartesian to Spherical and Cylindrical polar coordinates). Applications: Areas by double integrals and volumes by triple integrals.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Edition, Narosa Publications, 2016

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 13th Edition, John Wiley & Sons, 2006
2. Thomas' Calculus, G. B. Thomas, 13th Edition, Pearson, 2014
3. A Text Book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS1PH101) APPLIED PHYSICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: 10 + 2 Physics

COURSE OBJECTIVES:

- To explore the working and applications of lasers and fibre optics in modern technology
- To study crystal structures and defects in materials
- To learn nano materials and characterization techniques like XRD & SEM
- To understand the properties and applications of dielectric and magnetic materials
- To identify the importance of quantum mechanics and quantum computing

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Explain the principles of lasers and fibre optics and their applications in communication and sensing

CO-2: Understand different types of crystals and the importance of crystal defects

CO-3: Identify the role of nanomaterials in science and engineering applications

CO-4: Classify the dielectric and magnetic materials to illustrate their applications

CO-5: Realize the role of quantum mechanics in quantum computing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	2	1	-	-	-	1	2	-
CO-2	3	3	-	-	-	-	1	-	-	-	1	1	-
CO-3	3	2	2	1	1	2	1	-	-	-	1	-	-
CO-4	3	3	1	1	1	2	1	-	-	-	1	1	-
CO-5	3	3	1	1	1	2	1	-	-	-	1	-	-

UNIT-I:

Laser and Fibre Optics: Introduction to laser, characteristics of laser, Einstein coefficients and their relations, metastable state, population inversion, pumping, lasing action, Ruby laser, He-Ne laser, Semiconductor laser, applications: Bar code scanner, LIDAR for autonomous vehicle.

Introduction to fibre optics, total internal reflection, construction of optical fibre, acceptance angle, numerical aperture, classification of optical fibres, losses in optical fibre, applications: optical fibre for communication system, sensor for structural health monitoring.

UNIT-II:

Crystallography & Defects: Introduction: Unit cell, space lattice, basis, lattice parameters; crystal structures, Bravais lattices, packing factor: SC, BCC, FCC; Miller indices, inter-planar distance; Introduction to defects, Point defects (Vacancies, Interstitial and Impurities) Line imperfections, Edge and Screw dislocation, Burger vector, Surface defects and volume defects (Qualitative Treatment).

UNIT-III:

Nano Materials & Characterization: Concept of Nanomaterials: surface to volume ratio and quantum confinement, Fabrication: Sol-Gel, Ball milling, Applications. **Characterization:** X -ray diffraction, Bragg's law, powder method, Calculation of average crystallite size using Debye Scherrer's formula, scanning electron microscopy (SEM): block diagram, working principle.

UNIT-IV:

Dielectric and Magnetic Materials: Introduction to dielectric materials, types of polarization (qualitative): electronics, ionic & orientation; ferroelectric, piezoelectric, pyroelectric materials and their applications: Ferroelectric Random-Access Memory (Fe-RAM), load cell and fire sensor.

Introduction to magnetic materials, classification of magnetic materials, hysteresis, Weiss domain theory of ferromagnetism, soft and hard magnetic materials, applications: magnetic hyperthermia for cancer treatment, magnets for EV, Giant Magneto Resistance (GMR) device.

UNIT-V:

Quantum Mechanics and Quantum Computing: Introduction, de-Broglie hypothesis, Heisenberg uncertainty principle, physical significance of wave function, postulates of quantum mechanics: operators in quantum mechanics, eigen values and eigen functions, expectation value; Schrödinger's time independent wave equation, particle in a 1D box.

Introduction, concept of quantum computer, advantages of quantum computing over classical computation. classical bits, Qubits, multiple Qubit system, quantum computing system for information processing, Dirac's Bra and Ket notation and their properties, Introduction to entanglement.

TEXT BOOKS:

1. Textbook of Engineering Physics, M. N. Avadhanulu, P. G. Kshirsagar & T. V. S. Arun Murthy, 11th Edition, S. Chand, 2019
2. Introduction to Solid State Physics, Charles Kittel, John Wiley & Sons
3. Introduction to Classical and Quantum Computing, Thomas G. Wong, Rooted Grove

REFERENCES:

1. Engineering Physics, B. K. Pandey and S. Chaturvedi, 2nd Edition, Cengage Learning, 2022
2. Energy Materials a Short Introduction to Functional Materials for Energy Conversion and Storage, A. S. Bandarenka, CRC Press, Taylor & Francis Group Energy Materials, Taylor & Francis Group, 1st Edition, 2022
3. Concepts of Modern Physics, Aurther Beiser, McGraw-Hill

4. Engineering Physics, P. K. Palanisamy, SciTech Publications
5. Quantum Computing, Jozef Gruska, McGraw-Hill

ONLINE RESOURCES:

1. <https://shijuinpallotti.wordpress.com/wp-content/uploads/2019/07/optical-fiber-communications-principles-and-pr.pdf>
2. https://www.geokniga.org/bookfiles/geokniga-crystallography_0.pdf
3. <https://dpbck.ac.in/wp-content/uploads/2022/10/Introduction-to-Solid-State-PhysicsCharles-Kittel.pdf>
4. <https://www.thomaswong.net/introduction-to-classical-and-quantum-computing-1e4p.pdf>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES1DS101) PROBLEM SOLVING THROUGH C

(Common to AME and Bio-Technology)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To relate basics of programming language constructs and problem-solving techniques
- To classify and implement control structures and derived data types
- To analyze and develop effective modular programming
- To construct mathematical problems and real time applications using C language

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understanding computer fundamentals, problem-solving tools and implement programs using various data types, operators, and expressions

CO-2: Apply conditional and iterative control structures to solve computational problems

CO-3: Design and implement programs using arrays (1D, 2D) and strings

CO-4: Apply modular programming principles using functions and perform basic file handling operations

CO-5: Implement advanced C constructs for solving real-world problems

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	3	2	1	-	-	-	2	2	-	2	-
CO-2	2	2	3	2	1	1	-	-	2	2	-	2	-
CO-3	1	2	3	1	1	1	-	-	2	2	2	2	-
CO-4	1	2	3	1	1	1	-	-	2	2	2	2	-
CO-5	2	2	2	2	1	1	-	-	2	2	2	2	-

UNIT-I:

Introduction to Programming: Introduction to components of a computer system (I/O devices, memory, processor). Algorithm, Flow chart with examples. Introduction to the C Language – Programming Language levels, Identifiers, Data Types, Variables, Constants, Input / Output, Operators (Arithmetic, relational, logical, bitwise etc.), Expressions, Precedence and Associativity, Type casting and Type conversion, Simple C Program examples.

UNIT-II:

Conditional Branching and Loops: Statements- Selection Statements (making decisions) – branching with if, if-else, nested if-else, else-if ladder, switch case

Repetition statements (loops)-while, for, do-while statements, nested loops, other statements related to looping– break, continue.

UNIT-III:

Arrays: One- and two-dimensional arrays, creating, accessing and manipulating elements of arrays.

Strings: Introduction to strings, handling strings as array of characters, string and character functions available in C, arrays of strings.

UNIT-IV:

Functions: Declaring a function, function types with respect to return values and parameters, call by value, call by reference. Recursion with examples. Some C standard functions and libraries, Storage classes.

UNIT-V:

Pointers: Idea of pointers, defining pointers, use of pointers in self-referential structures, dynamic memory allocation.

Structures & Unions: Defining structures and array of structures, Unions.

Files: File types, character wise reading and writing operations (fopen(), fclose(), fputc(), fgetc()).

TEXT BOOKS:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India
2. Schaum's Outline of Programming with C, Byron Gottfried, McGraw-Hill

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, IV Edition, McGraw-Hill
2. Let Us C, Yashvant Kanetkar, BPB Publications
3. Programming in ANSI C, E. Balaguruswamy, Tata McGraw-Hill

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25PC1AE101) MANUFACTURING TECHNOLOGY

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To understand the various metal casting processes
- To impart the knowledge of various welding processes
- To understand the importance of mechanical working
- To learn various sheet metal working processes
- To appreciate various plastic manufacturing processes

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Select the suitable casting technique for manufacturing of the components

CO-2: Recommend suitable welding processes for different applications

CO-3: Summarize the various mechanical working processes

CO-4: Choose suitable sheet metal operations

CO-5: Demonstrate various plastic manufacturing processes

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	1	1	1	1	1	1	2	3	1	2	3	2
CO-2	3	1	1	1	1	1	1	2	3	1	2	3	2
CO-3	3	1	1	1	1	1	1	2	3	1	2	3	2
CO-4	3	1	1	1	1	1	1	2	3	1	2	3	2
CO-5	3	1	1	1	1	1	1	2	3	1	2	3	2

UNIT-I:

Introduction: Manufacturing process, Classification.

Casting: Steps involved in making a casting, Advantage of casting and its applications, Types of patterns – Materials used for patterns, Pattern allowances, Types of foundry sands.

{Limited to processes, advantages, disadvantages and applications only}

Casting Processes: Centrifugal casting, Die casting, Investment casting, shell moulding– Advantages and disadvantages; Applications, Casting defects.

Furnaces: Cupola furnace and Electric arc furnace.

{Limited to processes, advantages, disadvantages and applications only}.

UNIT-II:

Welding: Classification of welding processes, types of welded joints, Arc welding, Gas welding, TIG and MIG welding, Resistance welding, Thermit welding, Friction stir welding, Soldering and Brazing, Welding defects.

{Limited to processes, advantages, disadvantages and applications only}

UNIT-III:

Mechanical Working Processes: Introduction, Recrystallisation, Hot working, Cold working.

Rolling: Principle, Rolling stand arrangement, Rolling defects

Forging Processes - Principles of forging, Tools and dies, Types of Forging, Smith forging, Drop Forging, Forging defects.

Extrusion: Hot extrusion and Cold extrusion, Forward extrusion and Backward extrusion, Impact extrusion, Hydrostatic extrusion, Extrusion defects.

{Limited to processes, advantages, disadvantages and applications only}.

UNIT-IV:

Sheet Metal Operations:

Press Tool Operations: Classification, Shearing action; Blanking, piercing Trimming, Shaving, Nibbling, notching, lancing; Bending and forming, Drawing and its types: Deep drawing, Wire drawing and Tube drawing, Embossing and Coining, Spinning, Bending and its types

{Limited to processes, advantages, disadvantages and applications only}

UNIT-V:

Processing of Plastics: Types of Plastics, Properties, Advantages, Disadvantages and Applications.

Processing Methods: Blow molding, injection molding, compression molding, transfer molding.

{Limited to processes, advantages, disadvantages and applications only}

TEXT BOOKS:

1. Manufacturing Technology, P. N. Rao, Volume - I, 5th Edition, McGraw-Hill, 2018
2. Production Technology, R. K. Jain, Khanna Publishers, 2004

REFERENCES:

1. Elements of Workshop Technology S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, Vol. 1, 13th Edition, Media Promoters & Publishers, 2010
2. Manufacturing Processes, H. N. Gupta, R. C. Gupta, Arun Mittal, 2nd Edition, New Age International, 2009
3. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Publishers, 2001
4. A Text Book of Production Technology, P. C. Sharma, 8th Edition, S. Chand, 2014
5. Principles of Modern Manufacturing, Mikell P. Groover, 5th Edition, Wiley, 2014

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/112107144>
2. <https://nptel.ac.in/courses/112104195>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25HS1EN101) ENGLISH FOR SKILL ENHANCEMENT

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Basic English

COURSE OBJECTIVES:

- To enhance vocabulary through word formation processes
- To read and comprehend different kinds of texts
- To write clear, concise paragraphs, essays, and letters to produce appropriate technical prose
- To improve coherence and cohesion in writing and speaking
- To recognize and practice the use of rhetorical elements necessary for the successful practice of scientific and technical communication

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Use vocabulary contextually and effectively in oral and written communication

CO-2: Demonstrate an understanding of the rules of functional Remedial Grammar and sentence structures

CO-3: Demonstrate improved competence in Standard Written English, including Remedial Grammar, sentence and paragraph structure, and coherence, and use this knowledge to accurately communicate technical information

CO-4: Apply principles of writing while developing paragraphs, essays, letters and technical prose

CO-5: Employ appropriate rhetorical patterns of discourse in scientific and technical communication

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	1	1	1	-	1	1	1	2	3	2	-	-
CO-2	2	2	2	2	2	2	2	1	3	3	2	-	-
CO-3	2	2	2	2	1	3	2	1	3	3	2	-	-
CO-4	1	1	1	1	1	2	2	1	2	3	2	-	-
CO-5	1	1	1	1	-	2	1	1	2	2	1	-	-

UNIT-I:

1. **Reading:** -The Generation Gap' by Benjamin M. Spock
The Lotos Eaters – Alfred Tennyson, The Land of High Passes – Bhagyalakshmi Krishnamurthy
2. **Grammar:** Degrees of comparisons, Conjunctions, and Prepositions – Identifying Common Errors

3. **Vocabulary:** Word Formation (Affixation, Compounding, Conversion, Blending, Borrowing)
4. **Writing:** Punctuation, Clauses and Sentences, Transitional Devices, Paragraph Writing- Process

UNIT-II:

1. **Reading:** Emerging Technologies
Medicate – Oleksandr Gorpynich Of Studies – Francis Bacon
2. **Grammar:** Articles Noun-Pronoun Agreement; Concord
3. **Vocabulary:** Word Formation (Prefixes, Suffixes, Root Words)
4. **Writing:** Essay Writing -Descriptive, Argumentative, Expository

UNIT-III:

1. **Reading: Leisure- William Henry Davies -Be Thankful - Unknown Author**
The Rising of the Moon – Lady Gregory, With The Photographer – Stephen Leacock
2. **Grammar:** Misplaced Modifiers
3. **Vocabulary:** Synonyms and Antonyms
4. **Writing:** Letter Writing- Formal Letter Writing– Letter of Complaint, Letter of Requisition, Email Writing, Email Etiquette

UNIT-IV:

1. **Reading:** -Why a Start-Up Needs to Find its Customers First'- Pranav Jain
Too dear – Leo Tolstoy, Shooting an Elephant – George Orwell
2. **Grammar:** Clichés, Redundancies
3. **Vocabulary:** Common Abbreviations
4. **Writing:** Summary Writing, Job Application, Resume Writing

UNIT-V:

1. Reading: '**Professional Ethics**'
2. Patterns of Writing: Comparison and Contrast
3. Patterns of Writing: Cause and Effect Paragraphs
4. Patterns of Writing: Classification Paragraph
5. Patterns of Writing: Problem–Solution Pattern of Writing

TEXT BOOKS:

1. Board of Editors, English for the Young in the Digital World, Orient Black Swan, 2025
2. Unlocking English: A Skill-Based Approach for Language Proficiency
3. Technical Communication, Rebecca E. Burnett, 6th Edition, Cengage Learning

REFERENCES:

1. Practical English Usage, Swan Michael, New Edition, Oxford University Press, 2016
2. English Grammar Just for You, Karal, Rajeevan, Oxford University Press, 2023
3. Empowering with Language: Communicative English for Undergraduates, Cengage Learning, 2024
4. Communication Skills – A Workbook, Sanjay Kumar & Pushp Lata, Oxford University Press, 2022
5. Study Writing, Hamp Liz., Lyons and Heasley, Ben, C U P, 2006

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25BS2PH101) APPLIED PHYSICS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To explore the working principle of lasers and optical fibers through hands-on experiments
- To analyze the characteristics of semiconductors and devices
- To study resonance and magnetic induction in electromagnetic systems
- To synthesize the magnetic materials and study magnetic, dielectric properties of materials
- To develop skills in data analysis, interpretation, and scientific reporting

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Verify laser and optical fibre parameters through experimental study

CO-2: Characterize semiconductors and devices such as solar cell, laser diode etc

CO-3: Demonstrate resonance phenomena and realize tangent law of magnetism

CO-4: Understand the magnetic and dielectric properties of materials

CO-5: Apply scientific methods for accurate data collection, analysis, and technical report writing

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-2	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-3	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-4	3	3	1	2	1	2	1	3	2	-	1	1	-
CO-5	3	3	1	2	2	2	1	3	3	2	1	-	2

LIST OF EXPERIMENTS:

(Any 8 experiments are to be performed)

- Determination of wavelength of a laser using diffraction grating.
 - Study of V-I & L-I characteristics of a given laser diode.
- Determination of numerical aperture of a given optical fibre
 - Determination of bending losses of a given optical fibre.
- Synthesis of magnetite (Fe₃O₄) powder using sol-gel method.
- V-I characteristics of solar cell.
- Magnetic field along the axis of current carrying coil – Stewart and Gee

6. Melde's Experiment.
7. Study of B-H curve of a ferro magnetic material.
8. Study of P-E loop of given ferroelectric crystal.
9. Determination of dielectric constant of a given material.
10. Determination of frequency of AC mains using sonometer.

TEXT BOOKS:

1. Engineering Physics Laboratory Manual/Observation, Physics Faculty, VNRVJIET
2. A Textbook of Practical Physics, S. Balasubramanian, M. N. Srinivasan, S. Chand Publishers, 2017

REFERENCES:

1. Engineering Physics Lab Manual, Dr. Y. Aparna & Dr. K. Venkateswarao, VGS Book
2. Engineering Physics Practicals, Dr. B. Srinivasa Rao, V. K. V. Krishna, K. S. Rudramamba, University Science Press, Imprint of Laxmi Publications

ONLINE RESOURCES:

1. <https://vlab.amrita.edu/index.php?sub=1&brch=189&sim=343&cnt=1>
2. <https://vlab.amrita.edu/index.php?sub=1&brch=280&sim=1518&cnt=1>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2DS101) PROBLEM SOLVING THROUGH C LABORATORY

(Common to AME and Bio-Technology)

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To acquire hands-on experience in problem solving using C language
- To develop modular, efficient and readable programs using control structures, arrays, strings, and functions
- To learn concepts of recursion, pointers, structures, and unions
- To strengthen problem-solving skills with file handling, dynamic memory allocation

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply data types, operators, and control structures to solve computational problems

CO-2: Develop modular programs using functions and recursion

CO-3: Implement programs using arrays, strings, pointers, structures, and unions

CO-4: Solve real-time problems using file handling and dynamic memory concepts

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	3	2	1	-	-	-	2	-	-	1	-
CO-2	2	2	3	2	1	1	-	-	2	2	2	1	-
CO-3	1	2	3	1	1	1	-	-	2	2	2	1	-
CO-4	2	1	3	1	1	1	-	-	2	2	2	1	-

LIST OF EXPERIMENTS:

WEEK 1:

- Programs on input/output statements (scanf, printf).
- Programs on arithmetic operators (sum, difference, product, quotient).
- Programs on expression evaluation, operator precedence, type conversion, and casting.

WEEK 2:

- Program to check whether a number is even or odd using if-else.
- Program to find the largest of three numbers using nested if-else.
- Program to implement a simple calculator using switch-case.

WEEK 3:

- Program to print multiplication table of a given number using for.
- Program to display sum of first N natural numbers using while.
- Program to compute factorial of a number using do-while.

WEEK 4:

- Program to print patterns (triangles, pyramids) using nested loops.
- Program to demonstrate break, continue, and goto.

WEEK 5:

- Program to read and find sum and average of elements in an array.
- Program to search for an element using linear and binary search.
- Program to sort elements using bubble, selection, and insertion sort.

WEEK 6:

- Program to perform addition of two matrices.
- Program to multiply two matrices.
- Program to find the transpose of a matrix.

WEEK 7:

- Program to check whether a given string is palindrome.
- Program to search for a substring in a given string.
- Programs using string handling functions (strlen, strcpy, strcmp, strcat).

WEEK 8:

- Lab Internal – I

WEEK 9:

- Program to swap two numbers using call by value and call by reference.
- Program to generate Fibonacci series using recursion.
- Program to solve Tower of Hanoi problem using recursion.

WEEK 10:

- Program to store and display student details using structures.
- Program using array of structures (e. g., employee records).
- Program using unions (demonstrating memory sharing).

WEEK 11:

- Program to demonstrate pointers to variables and arrays (1-D and 2-D).
- Program using pointers to strings.

WEEK 12:

- Program to allocate memory dynamically using malloc, calloc, and release with free.

WEEK 13:

- Program to write student records into a file.
- Program to read and display student records from a file.

WEEK 14:

- Lab Internal – II

TEXT BOOKS:

1. The C Programming Language, Brian W. Kernighan and Dennis M. Ritchie, Prentice Hall of India
2. Schaum's Outline of Programming with C, Byron Gottfried, McGraw-Hill

REFERENCES:

1. C: The Complete Reference, Herbert Schildt, 4th Edition, McGraw-Hill
2. Let Us C, Yashvant Kanetkar, BPB Publications
3. Programming in ANSI C, E. Balaguruswamy, Tata McGraw-Hill

B. Tech. I Semester

(25HS2EN101) ENGLISH LANGUAGE AND COMMUNICATION SKILLS LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

INTRODUCTION:

The aim of the English Language and Communication Skills Lab (ELCS Lab) is to develop the communicative skills of learners to function productively in academic and professional settings. This course focuses on developing Listening Skills, Reading Skills, Speaking Skills, Information transfer, Vocabulary, and pronunciation.

Professional students need reading strategies to explore scientific information and process the information. Further, they need to present and share this information in speaking and/or writing with their peers and scientific community. In addition to these, they also need good vocabulary for the presentation of their ideas. Practice is given not only in processing information to identify main ideas and relevant facts, to compare and contrast information and finally draw conclusions but also in presenting the same in speaking and writing.

COURSE OBJECTIVES:

- To train students to use neutral accent through phonetic sounds, symbols, stress and intonation
- To provide practice in vocabulary usage & grammatical construction
- To provide ample practice in LSRW skills and train the students in oral presentations, public speaking, role play, and situational dialogue
- To provide practice in defining technical terms and describing processes
- To equip students with excellent writing skills and information transfer skills

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Speak fluently with a neutral accent

CO-2: Use contextually apt vocabulary and sentence structures

CO-3: Make Presentations with great confidence

CO-4: Define technical terms and describe processes

CO-5: Write accurately, coherently, and lucidly

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	1	2	2	2	-	-	-	2	3	-	2	-	-
CO-2	1	2	2	2	3	3	2	2	3	2	2	-	-
CO-3	-	1	2	1	-	1	2	2	3	2	2	-	-
CO-4	2	2	2	2	2	-	-	1	3	2	2	-	-
CO-5	1	2	1	2	2	2	2	1	3	3	3	-	-

LIST OF EXPERIMENTS:

UNIT-I:

CALL

1. Common errors with Speech Sounds – Identify elements of Mother Tongue Influence,
2. Listening Skills – Types of Listening- Barriers – Active Listening Practice
3. Reading Skills -Skimming & Scanning-Reading for Gist -Reading for Specific Details

Oral Communication Lab:

1. Non-Verbal Communication
2. Formal and Informal English, Small Talk
3. Greetings – Introducing oneself & others - Elevator Pitch

UNIT-II:

CALL

1. **Activity:** Listening for information
2. Listening comprehension
3. **Reading Skills:** Reading comprehension-contextual vocabulary

Oral Communication Lab

1. Participatory Conversation
2. Role Playing in Formal Situations, Situational Dialogues – Two-Way (E. g. Elevator Pitch)
3. Media Etiquette, Conference Etiquette

UNIT-III:

CALL

1. **Phonetics:** Errors in Pronunciation, Neutralizing Mother Tongue Influence
2. **Listening Skills:** Listening for Gist
3. **Reading Skills:** SQ3R Method, Inferring

Oral Communication Lab

1. Describing Objects / Situations/ Places, People, Events,
2. Process Description

UNIT-IV:

CALL

1. **Listening Skills:** Techniques for Listening –Listening for Specific Details
2. **Reading Skills:** Making inferences

Oral Communication Lab

1. Story Telling; Story STAR, Sequencing, Creativity
2. Telling and Retelling Stories- Collage

UNIT-V:

CALL

1. **Listening Skills:** Listening for Evaluation, Writing the Summary
2. **Prompt Engineering:** Generating Appropriate Prompts, Keyword Extraction
3. Definition of a Technical Term; System Flow

Oral Communication Lab

1. Report Presentation
2. Short Presentation

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. I Semester

(25ES2ME101) ENGINEERING WORKSHOP

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To know the different popular manufacturing process
- To gain a good basic working knowledge required for the production of various engineering products
- To provide hands on experience about use of different engineering materials, tools, equipment and processes those are common in the engineering field
- To identify and use marking out tools, hand tools, measuring equipment and to work to prescribed tolerances

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand various types of manufacturing processes

CO-2: Fabricate/make components from wood and steels through hands on experience

CO-3: Understand different machining processes like turning, drilling, tapping, etc.

CO-4: Understand electrical and electronic components and their assembly

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-2	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-3	-	-	1	-	-	-	-	3	3	2	-	2	1
CO-4	-	-	1	-	-	-	-	3	3	2	-	1	1

LIST OF EXPERIMENTS:

1. TRADES FOR EXERCISES:

- Carpentry:** Cross lap joint, Mortise & tenon joint
- Fitting:** L-fitting, Square fitting
- Tin smithy:** Rectangular tray, Rectangular scoop
- Arc welding:** Lap joint, Butt joint
- House Wiring:** Single lamp connection, Staircase connection
- Black smithy:** Round to Square, U-Hook
- Machine shop:** Step turning on lathe, Drilling & tapping

2. TRADES FOR DEMONSTRATION:

Plumbing, 3D printing, Power tools in construction and wood working.

TEXT BOOKS:

1. Workshop Manual, P. Kannaiah and K. L. Narayana, 3rd Edition, Scitech, 2015
2. Elements of Workshop Technology Vol. 1 & 2, S. K. Hajra Choudhury, A. K. Hajra Choudhury and Nirjhar Roy, 13th Edition, Media Promoters & Publishers, 2010

REFERENCES:

1. Manufacturing Engineering and Technology, Serope Kalpakjian, Steven R. Schmid, 4th Edition, Pearson Education, 2002
2. Manufacturing Technology-I, S. Gowri, P. Hariharan and A. Suresh Babu, Pearson Education, 2008
3. Processes and Materials of Manufacture, Roy A. Lindberg, 4th Edition, Prentice Hall India, 1998
4. Manufacturing Technology Vol-1 & 2, P. N. Rao, Tata McGraw-Hill, 2017

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1MT105) ORDINARY DIFFERENTIAL EQUATIONS AND VECTOR CALCULUS

(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
2	1	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematical Knowledge at Pre-University level

COURSE OBJECTIVES:

- To learn methods of solving first order differential equations and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with constant coefficients and learn about its applications to basic engineering problems.
- To learn methods of solving higher order differential equations with variable coefficients and learn about its applications to basic engineering problems.
- To learn the physical quantities involved in engineering field related to vector valued functions
- To know the basic properties of vector valued functions and their applications to line, surface and volume integrals

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Formulate and solve the problems of first order differential equations

CO-2: Solve higher order differential equation with constant coefficients and apply the concept of differential equation to real world problems

CO-3: Solve the higher order differential equation with variable coefficients and apply the concept of differential equation to real world problems

CO-4: Find the gradient, divergence, curl and its physical interpretations

CO-5: Evaluate the line, surface and volume integrals and converting them from one to another

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	1	2	1	-	-	-	-	-	-	2	1
CO-2	3	3	1	2	1	--	-	-	-	-	-	2	1
CO-3	3	2	1	2	2	-	-	-	-	-	-	2	2
CO-4	3	2	2	2	1	-	-	-	-	-	-	1	2
CO-5	3	2	2	3	1	-	-	-	-	-	-	2	1

UNIT-I:

First Order Ordinary Differential Equations: Exact differential equations – Equations reducible to exact differential equations – linear and Bernoulli's equations – Orthogonal Trajectories (only in Cartesian Coordinates). Applications: Newton's law of cooling – Law of natural growth and decay.

UNIT-II:

Higher Order ODEs with Constant Coefficients: Higher order linear differential equations with constant coefficients: Non-Homogeneous terms of the type e^{ax} , $\sin ax$, $\cos ax$, polynomials in x , $e^{ax} V(x)$ and $x V(x)$.

UNIT-III:

Higher Order ODEs with Variable Coefficients: Higher order linear Ordinary differential equations with variable coefficients, Method of variation of parameters, Euler-Cauchy ODEs, Legendre's ODEs.

UNIT-IV:

Vector Differentiation: Vector point functions and scalar point functions – Gradient – Directional derivatives -Divergence and Curl - Solenoidal and Irrotational vectors - Scalar potential functions – Vector Identities (without proof).

UNIT-V:

Vector Integration: Line, Surface and Volume Integrals. Theorems of Green, Gauss and Stokes (without proofs) and their applications.

TEXT BOOKS:

1. Higher Engineering Mathematics, B. S. Grewal, 36th Edition, Khanna Publishers, 2010
2. Advanced Engineering Mathematics, R. K. Jain and S. R. K. Iyengar, 5th Edition, Narosa Publications, 2016
3. Higher Engineering Mathematics, B. V. Ramana, 11th Reprint, Tata McGraw-Hill, 2010

REFERENCES:

1. Advanced Engineering Mathematics, Erwin Kreyszig, 9th Edition, John Wiley & Sons, 2006
2. Advanced Engineering Mathematics, Dennis G. Zill, 6th Edition
3. A Text Book of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, 2008
4. Higher Engineering Mathematics, H. K. Dass and Er. Rajnish Verma, S. Chand

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS1CH102) ENGINEERING CHEMISTRY

(Common to CE, ME and AE)

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: General Chemistry and Basic Mathematics

COURSE OBJECTIVES:

- To analyze the quality of water for sustainable living
- To imbibe the conceptual knowledge of electrochemical processes and corrosion science
- To outline the importance of non-conventional energy sources and portable electric devices
- To acquire the knowledge about polymer science and its applications in various fields
- To recognize the significance of analytical techniques and materials in engineering, industrial, environmental, and biomedical applications

COURSE OUTCOMES: After completion of the course, the student should be able to
CO-1: Assess the specification of water regarding its usage in domestic & Industrial scenarios

CO-2: Interpret electrode potentials and predict the suitable corrosion control methods in safeguarding the structures

CO-3: Recognize the transformations in energy sources & battery technology

CO-4: Analyze the efficacy of polymers in diverse applications

CO-5: Identify the principles of UV-Visible and IR spectroscopy and materials in varied applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	2	2	2	2	1	1	1	1	2	-	1
CO-2	3	2	2	1	1	2	1	1	1	1	2	-	1
CO-3	3	2	2	1	1	1	1	1	1	1	2	-	1
CO-4	3	2	2	1	1	2	1	1	1	1	2	-	1
CO-5	3	2	1	1	1	1	1	1	1	1	2	-	1

UNIT-I:

Water and its Treatment: Introduction – Hardness, types, degree of hardness and units – numerical Problems. Estimation of hardness of water by complexometric method. Potable water and its specifications (WHO) – Steps involved in the treatment of potable water – screening, sedimentation, coagulation, filtration and disinfection by chlorination and break-point chlorination.

Boiler Troubles: Scales, Sludges, Internal treatment of boiler feed water – Calgon conditioning, Phosphate conditioning, Colloidal conditioning. External treatment methods – Softening of water by Ion-exchange process. Membrane separation – Types of membranes [Microfiltration (MF), Ultrafiltration (UF), Nanofiltration (NF)], advantages and applications. Desalination of brackish water – Reverse osmosis.

UNIT-II:

Electrochemistry and Corrosion: Introduction – Electrochemical cell & galvanic cell, electrode potential, standard electrode potential, Nernst equation (no derivation). Types of electrodes – Reference electrodes – Construction, working principle, advantages, limitations and applications of Primary reference electrode (Standard Hydrogen Electrode – SHE), Secondary reference electrode (Calomel electrode) and Ion-selective electrode (Combined glass electrode).

Corrosion: Introduction – Definition, causes and effects of corrosion, Theories of corrosion – chemical and electrochemical theories of corrosion. Types of corrosion – galvanic, waterline and pitting corrosion. Factors affecting rate of corrosion – nature of metal (position, passivity, purity, areas of anode and cathode) & nature of environment (temperature, pH, humidity). Corrosion control methods – Cathodic protection Method – Sacrificial anode and impressed current cathodic protection methods.

UNIT-III:

Energy Sources and Battery Technology:

Fuels: Introduction and characteristics of good fuel, classification based on physical state.

Fossil Fuels: Petroleum – Refining of Crude oil, Cracking – Types of cracking – Moving bed catalytic cracking. Synthetic Fuels: Fischer-Tropsch process.

Gaseous Fuels: Composition and uses of LPG, CNG and Hythane. Green Hydrogen (generation by water electrolysis – PEM, Alkaline, Solid Oxide).

Batteries: Introduction – Classification of batteries – Primary, secondary, reserve batteries and fuel cells (differences and examples). Construction, working and applications of Zn-air (primary battery) and Lithium-ion battery (secondary battery), Characteristic features and Advantages of Metal ion batteries. (Sodium Ion Battery). Fuel Cells – Construction and applications of Proton Exchange Membrane Fuel Cell (PEMFC).

UNIT-IV:

Polymers: Definition – Classification of polymers – Types of polymerizations – Addition and condensation polymerization, differences between thermoplastics and thermosetting plastics. Plastics, Elastomers and Fibers: Synthesis, properties and applications of Teflon, Buna-S and Kevlar.

Fiber Reinforced Plastics (FRPs): Classification, features and applications.

Conducting Polymers: Definition, classification and applications.

Shape Memory Polymers: Definition, classification and applications.

Biodegradable polymers: Introduction – Definition features & applications of Polylactic acid (PLA).

UNIT-V:

Spectroscopic Applications and Engineering Materials: Interpretative spectroscopic applications of UV-Visible spectroscopy for analysis of pollutants in dye industry, IR spectroscopy in night vision – security, Pollution Under Control – CO sensor (Passive Infrared detection).

Self-healing Materials: Definition, working principle and applications (self-healing concrete).

Nanomaterials: Characterization by STM & AFM.

Lubricants: Definition and characteristics of a good lubricant – thin film mechanism of lubrication, properties of lubricants – viscosity, cloud and pour point, flash and fire point.

TEXT BOOKS:

1. Engineering Chemistry, P. C. Jain and M. Jain, Dhanpat Rai and Company, 2010
2. Engineering Chemistry, Rama Devi, Dr. P. Aparna and Rath, Cengage learning, 2025
3. Engineering Chemistry, Thirumala Chary, Laxminarayana & Shashikala, Pearson Publications, 2020

REFERENCES:

1. Engineering Chemistry, Shashi Chawla, Dhanpat Rai and Company, 2011
2. Engineering Chemistry, Shikha Agarwal, Cambridge University Press, 2015
3. Textbook of Engineering Chemistry, Jaya Shree Anireddy, Wiley Publications
4. Engineering Analysis of Smart Material Systems, Donald J. Leo, Wiley, 2007
5. Challenges and Opportunities in Green Hydrogen, Paramvir Singh, Avinash Kumar Agarwal, Anupma Thakur, R. K. Sinha, Springer Nature, 2024

B. Tech. II Semester

(25ES1DS102) DATA STRUCTURES THROUGH C

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE OBJECTIVES:

- To introduce various searching and sorting techniques
- To demonstrate operations on linear and non-linear data structure
- To develop an application using suitable data structure
- To efficiently implement the different data structures and solutions for specific problems

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Understand basic concepts of data structures and Searching techniques

CO-2: Understand the dynamic data structures, like Linked List, Stacks and Queues

CO-3: Execute the given problem using non-linear data structures

CO-4: Apply sorting Techniques on Arrays

CO-5: Analyze graph concepts in real world scenarios

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	-	-	3	-	-	-	-	2	-	1	-
CO-2	3	3	2	2	3	-	-	-	2	2	-	1	-
CO-3	3	3	2	2	3	-	-	-	2	2	-	1	-
CO-4	3	3	2	2	3	-	-	-	2	2	-	1	-
CO-5	3	3	3	2	3	2	2	2	2	2	1	1	-

UNIT-I:

Introduction to Data Structures: Abstract Data Types (ADT) Searching: Linear Search and Binary Search Techniques.

Linear Data Structures: Stacks - ADT Stack and its operations: Applications of Stacks: Recursion, Expression Conversion and evaluation.

UNIT-II:

Linear Data Structures: Queues - ADT queue, Types of Queues: Linear Queue, Circular Queue, operations on each type of Queues.

Linked Lists: Singly linked lists: Representation in memory; Operations: Traversing, Searching, Insertion, Deletion from linked list; Linked representation of Stack and Queue.

Doubly Linked List: All operations

UNIT-III:

Trees: Basic Tree Terminologies, Different types of Trees: Binary Tree, Binary Search Tree, Tree traversal – Inorder, Preorder, Postorder and representations.

UNIT-IV:

Sorting: Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Heap Sort, Radix Sort.

UNIT-V:

Graphs: Graph terminology – Representation of graphs – Graph Traversal: BFS (breadth first search) – DFS (depth first search) – Minimum Spanning Tree.

TEXT BOOKS:

1. Fundamental of Data Structure, Horowitz and Sahani, Galgotia Publication
2. Data Structure, Lipschutz, Schaum Series
3. Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman

REFERENCES:

1. Algorithms, Data Structures, and Problem Solving with C++, Mark Allen Weiss, Addison-Wesley Publishing Company
2. How to Solve it by Computer, 2nd Impression, R. G. Dromey, Pearson Education
3. Introduction to Algorithms, Cormen, Leiserson and Rivest
4. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning
5. Data Structures using C & C++, Ten Baum, Prentice-Hall International

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/106102064>
2. https://onlinecourses.swayam2.ac.in/cec19_cs04/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1EE104) ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING

TEACHING SCHEME		
L	T/P	C
2	0	2

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To understand the use of electrical energy in different engineering fields
- To analyze electrical circuits using different network reduction techniques
- To practice the techniques to control, assess electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Appreciate the role of electrical energy in various engineering branches and analyze the DC circuits using various network reduction techniques

CO-2: Analyze the various electrical parameters of AC circuits with R-L-C elements

CO-3: Understand the operation of various Electrical Machines & Installations

CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	-	-	-	-	-	1	2	-
CO-2	2	2	1	1	-	-	-	-	-	-	1	2	-
CO-3	2	1	1	1	-	1	-	-	-	-	1	2	-
CO-4	2	1	1	1	-	-	-	-	-	-	1	2	-

UNIT-I:

Introduction to Electrical Energy and DC Circuits: The role of Electrical Energy in modern life and various engineering branches- Circuit Concept – Types of Elements R-L-C parameters – Voltage and Current sources – Independent and dependent sources- Kirchhoff's laws – network reduction techniques: series, parallel, series parallel, star/delta transformations, Superposition theorem

UNIT-II:

Steady State AC Circuits: Representation of sinusoidal waveforms, average and RMS values, form factor and peak factor, phasor representation, Analysis of single-phase AC circuits consisting of R, L, C, series RL, RC, RLC combinations, real power, reactive power, apparent power, power factor - Three-phase balanced circuits, voltage and current relations in star and delta connections (Derivation only).

UNIT-III:

Electrical Machines: Transformer principle, types, Equivalent circuit, Regulation and Efficiency, DC generator principle, types Emf equation, DC motor principle, types, Back emf, Speed control of shun motor. Three phase induction motor, types, principle, torque Slip characteristics, working principle of Synchronous generator, principle of operation of synchronous motor.

UNIT-IV:**Electrical Installations:**

Components of LT Switch Gear: SFU, MCB, ELCR, MCCB. Types of wiring, Need of earthing and types. Batteries-types-working and electrical characteristics of Lead-Acid battery, Elementary energy consumption calculations, Block diagram of UPS and its operation

UNIT-V:

Diode & Bipolar Junction Transistor: PN Junction Diode, Operation and Symbol, V-I Characteristics, Half Wave Rectifier, Full-Wave Rectifier, (Centre Tap and Bridge Rectifiers), Zener Diode, V-I Characteristics and applications, BJT, Construction (NPN and PNP transistors), Transistor configurations (CC, CB and CE), Transistor biasing, Transistor as an Amplifier.

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. <https://nptel.ac.in/courses/117106108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES1AE101) FUNDAMENTALS OF ENGINEERING MECHANICS

TEACHING SCHEME		
L	T/P	C
3	0	3

EVALUATION SCHEME				
SE	CA	ELA	SEE	TOTAL
30	05	05	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To understand the basic principles of engineering mechanics and their applications in analyzing force and moment systems
- To gain knowledge of friction and its role in practical applications
- To understand the importance of centroid, centre of gravity (mass)
- To compute the area moment of inertia and mass moment of inertia of standard and composite sections
- To apply kinematics and kinetics concepts for analyzing the motion of particles and rigid bodies

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply equilibrium laws to analyze force systems in engineering applications

CO-2: Solve problems of bodies subjected to friction

CO-3: Determine centroid and centre of gravity of regular and composite bodies

CO-4: Calculate area and mass moments of inertia of simple and composite sections

CO-5: Analyze kinematics and kinetics of particles using appropriate principles of mechanics

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	2	2	2	1	-	-	-	-	2	3	2
CO-2	3	3	2	2	2	1	-	-	-	-	2	3	2
CO-3	3	3	2	2	2	1	-	-	-	-	2	3	2
CO-4	3	3	2	2	2	1	-	-	-	-	2	3	2
CO-5	3	3	2	3	2	1	-	-	-	-	3	3	3

UNIT-I:

Introduction to Mechanics: Classification of Engineering Mechanics-Definitions-Classification of force system-Fundamental Laws of Mechanics-Lami's Theorem-Resolution and components of a force-Resolution of several concurrent forces-Equilibrium of a particle-Equilibrant-Equations of Equilibrium-Free body diagram-Moment of a force about a point-Varignon's theorem-Moment of a couple-Resolution of a force into a force and a couple-Coplanar Parallel force system.

UNIT-II:

Friction: Types of Friction-Laws of friction-Definitions: Coefficient of friction-Angle of friction-Angle of repose- Equilibrium of bodies on rough horizontal and inclined planes- Equilibrium of connected bodies on rough horizontal and inclined planes.

UNIT-III:

Centroid & Centre of Gravity: Introduction - Centroid - Centroid of lines- Standard area – Centroid of composite sections - Centre of gravity of bodies -theorems of pappus-guldinus.

UNIT-IV:

Area Moment of Inertia: Introduction-Second Moment of area-Radius of gyration- Parallel axis theorem-Perpendicular axis theorem- Polar moment of inertia- Moments of inertia of standard sections and composite sections.

Mass Moment of Inertia: Moment of Inertia of Masses – Transfer Formula for Mass moments of inertia – Mass moment of Inertia of bodies (rectangular plate, circular plate, uniform rod, cone)

UNIT-V:

Kinematics of Particles: Introduction-Equations of motion in a straight line under uniform acceleration-Rectilinear motion under variable acceleration-Projectiles-Terms used with the projectiles-Motion of body projected horizontally- Motion of body projected at an angle on level ground.

Kinetics of Particles: Newton's Second Law- D'Alembert's Principle and its applications in plane motion and connected bodies including pulleys.

TEXT BOOKS:

1. Engineering Mechanics, S. Timoshenko, D. H. Young & J. V. Rao, 5th Edition, McGraw-Hill Education, 2017
2. A Text Book of Engineering Mechanics, R. K. Bansal, 6th Edition, Laxmi Publications, 2013
3. Rajasekaran S and Sankarasubramanian G. Engineering Mechanics-Statics and Dynamics New Delhi: Vikas Publishing House, 2006

REFERENCES:

1. Engineering Mechanics: Statics and Dynamics, J. L. Meriam & L. G. Kraige, J. N. Bolton, 9th Edition, Wiley Publishers, 2021
2. Engineering Mechanics, R. C. Hibbeler, 14th Edition, Pearson Education, 2018
3. Engineering Mechanics, A. K. Tayal, 14th Edition, Umesh Publications, 2012
4. Engineering Mechanics, R. K. Rajput, 2nd Edition, Laxmi Publications, 2013
5. Singer's Engineering Mechanics, K. Vijaya Kumar Reddy & J. Suresh kumar, 3rd Edition, B. S. Publishers, 2011

ONLINE RESOURCES:

1. https://onlinecourses.nptel.ac.in/noc25_me108/preview

2. <https://www.udemy.com/course/engineering-mechanics-statics/?couponCode=PMNVD2025>
3. <https://www.udemy.com/course/myengineeringmechanics/?couponCode=PMNVD2025>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25ES3ME102) ENGINEERING GRAPHICS

TEACHING SCHEME		
L	T/P	C
0	6	3

EVALUATION SCHEME				
D-D	SE	CP	SEE	TOTAL
10	20	10	60	100

COURSE OBJECTIVES:

- To know the importance of engineering scales and curves
- To learn to use the orthographic projections for points, lines and planes in different positions
- To learn to draw the projections of solids and its sections in different positions
- To learn to draw the development of surfaces and intersection of solids
- To learn to draw the isometric projections of solids

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply the concepts of scales and engineering curves in construction using AutoCAD

CO-2: Solve the problem of projections of points, lines and planes in different positions using AutoCAD

CO-3: Draw the orthographic projections of solids and its sections in different positions using AutoCAD

CO-4: Obtain the development of surfaces of solids and intersection of solids using AutoCAD

CO-5: Construct the isometric projections of solids using AutoCAD

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-2	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-3	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-4	3	2	-	-	3	2	-	-	-	-	2	-	-
CO-5	3	2	-	-	3	2	-	-	-	-	2	-	-

Introduction to AutoCAD Software: User interface, Menu System, Toolbars (Draw, Modify, Dimension, Layers), Setting drawing area, Status Bar (ortho, grid, snap, osnap, iso, linewidth), Display control commands (pan, zoom), Print setup.

UNIT-I:

Introduction to Engineering Graphics: Principles of Engineering Graphics and their Significance, Conventions, Drawing instruments.

Engineering Curves

Conic Sections: Ellipse, Parabola and Hyperbola- General methods only, Rectangular Hyperbola

Cycloidal Curves & Involute: Cycloid, Epicycloid, Hypocycloid and Involute
Scales: Plain, Diagonal and Vernier Scales

UNIT-II:

Orthographic Projections: Principles of Orthographic Projections, Conventions, Projections of Points in all positions; Projections of lines and planes inclined to both the planes - Auxiliary Views

UNIT-III:

Projections of Regular Solids: Projections of regular Solids like Prism, Cylinder, Pyramid and Cone inclined to both the Planes - Auxiliary Views

Sections of Solids: Section and sectional views of right regular solids like Prism, Cylinder, Pyramid and Cone – Auxiliary Views

UNIT-IV:

Development of Surfaces of Right Regular Solids: Development of surfaces of Right Regular Solids like Prism, Pyramid, Cylinder and Cone

Intersection of Solids: Intersection of prism vs prism, cylinder vs cylinder

UNIT-V:

Isometric Projections: Principles of isometric projection, Isometric Scale, Isometric Views, Conventions, Isometric Views of lines, Planes and Solids like Prism, Pyramid, Cylinder and Cone

Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions.

TEXT BOOKS:

1. Engineering Drawing, N. D. Bhatt, 53rd Edition, Charotar Publishing House, 2016
2. Textbook on Engineering Drawing, K. L. Narayana & P. Kannaiah, Scitech Publishers, 2010
3. Engineering Drawing and Computer Graphics, M. B. Shah & B. C. Rana, Pearson Education, 2010

REFERENCES:

1. Engineering Drawing and Graphics using AutoCAD, T. Jeyapoovan, 3rd Edition, S. Chand and Company
2. Engineering Drawing, Basant Agrawal and C. M. Agrawal, 3rd Edition, McGraw-Hill
3. Mastering AutoCAD 2021 and AutoCAD LT 2021, George Omura and Brian C. Benton (AutoCAD 2021), 1st Edition, John Wiley & Sons, Indianapolis, Indiana

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/112103019>
2. http://vlabs.iitb.ac.in/vlabs-dev/labs/mit_bootcamp/egraphics_lab/labs/index.php

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B. Tech. II Semester

(25BS2CH101) ENGINEERING CHEMISTRY LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Basic Knowledge of Volumetric Analysis and Mathematics

COURSE OBJECTIVES:

- To understand the preparation of standard solutions and handling of instruments
- To determine and evaluate the water quality
- To measure physical properties like absorption of light, surface tension, pH, conductance, and viscosity of various liquids
- To conduct and collect the experimental data using different laboratory techniques
- To summarize the data and find the applicability to real world scenario

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Learn and apply the basic laboratory methodologies for the preparation of the standard solutions and handling of instruments

CO-2: Estimate the ions / metal ions present in domestic and industrial water

CO-3: Utilize the instrumental techniques to assess the physical properties of oils and water

CO-4: Analyze the experimental data to predict solutions for complex engineering problems

CO-5: Apply the skills gained to address societal issues related to real world scenario

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	2	1	1	-	1	1	1	1	1	2	-	-
CO-2	3	2	2	2	1	2	1	1	1	1	2	-	-
CO-3	3	2	1	2	1	2	1	1	1	1	2	1	1
CO-4	3	2	2	2	1	2	1	1	1	1	2	-	1
CO-5	3	2	2	2	1	2	1	1	1	1	2	-	1

LIST OF EXPERIMENTS:

VOLUMETRIC ANALYSIS:

- Estimation of hardness of water by complexometric method using EDTA.
- Determination of chloride content in the given sample water using Argentometric method.
- Determination of Iron using Potassium Dichromate.

COLORIMETRY:

4. Estimation of copper present in the given solution by colorimetric method.

CONDUCTOMETRY:

5. Conductometric titration of Acid vs Base.
6. Conductometric titration of mixture of strong acid and weak acid by strong base.

pH METRY:

7. Titration of Acid vs Base using pH metric method.
8. Determination of viscosity of sample oil by Redwood Viscometer-I.
9. Estimation of acid value of given lubricant oil.
10. Determination of surface tension of a liquid by drop method using Stalagmometer.

PREPARATIONS:

11. Preparation of Bakelite.
12. Preparation of Nylon-6,6.
13. Determination of rate of corrosion of mild steel in the presence and absence of inhibitor.

14. VIRTUAL LAB EXPERIMENTS / CBP:

- Construction of advanced Fuel cell and it's working.
- Smart materials for Biomedical applications
- Batteries for electrical vehicles.
- Functioning of solar cell (materials used) and its applications.

TEXT BOOKS:

1. Laboratory Manual on Engineering Chemistry, S. K. Bhasin and Sudha Rani, Dhanpat Rai Publications
2. College Practical Chemistry, V. K. Ahluwalia, Sunitha Dhingra, Adargh Gulati, University Press Pvt. Ltd.
3. Practical Chemistry, Dr. O. P. Pandey, D. N. Bajpai, and Dr. S. Giri, S. Chand

REFERENCES:

1. Vogel's Textbook of Quantitative Chemical Analysis, G. N. Jeffery, J. Bassett, J. Mendham and R. C. Denny, Longmann, ELBS
2. Advanced Practical Physical Chemistry, J. D. Yadav, Goel Publishing House
3. Practical Physical Chemistry, B. D. Khosla, R. Chand and Sons
4. Lab Manual for Engineering Chemistry, B. Ramadevi and P. Aparna, S. Chand, 2022
5. Vogel's Textbook of Practical Organic Chemistry, 5th Edition

B. Tech. II Semester

(25ES2DS102) DATA STRUCTURES THROUGH C LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE OBJECTIVES:

- To impart the basic concepts of data structures and algorithms
- To learn the concepts about searching and sorting
- To understand the basic concepts about stacks, queues, lists
- To know the concepts of trees and graphs

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Implement all operations on different linear data structures

CO-2: Develop all operations on different non-linear data Structures

CO-3: Apply various searching and sorting techniques

CO-4: Understand the complexity analysis of linear and nonlinear data structures

CO-5: Use appropriate data structure for any given problem

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	3	3	2	-	3	-	-	-	2	2	-	1	-
CO-2	3	3	2	2	3	-	-	-	2	2	-	1	-
CO-3	3	3	2	2	3	-	-	-	2	2	-	1	-
CO-4	2	2	1	2	-	-	-	-	2	2	-	1	-
CO-5	3	3	3	2	3	2	2	2	3	3	1	1	-

LIST OF EXPERIMENTS:

WEEK 1:

- a) Implement Linear Search b) Implement Binary Search

WEEK 2:

Implement the following

- a) Stack using Array b) Queues using Array

WEEK 3:

- a) Program to convert infix expression to postfix expression.
b) Program for postfix evaluation.

WEEK 4:

Implement the following

- a) Program for parenthesis matching.
- b) Circular Queue using Array

WEEK 5:

Implement Single Linked List Operations –

- a) Creation
- b) Insertion – Begin, Middle and End
- c) Deletion – Begin, Middle and End
- d) Display

WEEK 6:

Implement the following

- a) Stack using Linked List
- b) Queues using Linked List

WEEK 7:

Implement Double-Linked List Operations –

- a) Creation
- b) Insertion – Begin, Middle and End
- c) Deletion – Begin, Middle and End
- d) Display

WEEK 8:

Lab Internal - 1

WEEK 9:

Implement BST Traversals

- a) Creation
- b) Preorder
- c) Inorder
- d) Postorder

WEEK 10:

Implement following sorting techniques

- a) Bubble Sort b) Insertion Sort c) Selection Sort

WEEK 11:

Implement following sorting techniques

- a) Merge Sort b) Quick Sort

WEEK 12:

Implement following sorting techniques

- a) Heap Sort b) Radix Sort

WEEK 13:

Implement following Graph traversals

- a) BFS b) DFS

WEEK 14:

Internal Lab -2

TEXT BOOKS:

1. Fundamental of Data Structure, Horowitz and Sahani, Galgotia Publication

2. Data Structure, Lipschutz, Schaum Series
3. Data Structures and Algorithms, Alfred V. Aho, John E. Hopcroft, Jeffrey D. Ullman

REFERENCES

1. Algorithms, Data Structures, and Problem Solving with C++, Mark Allen Weiss, Addison-Wesley
2. How to Solve it by Computer, 2nd Impression, R. G. Dromey, Pearson Education
3. Introduction to Algorithms, Cormen, Leiserson and Rivest
4. Data Structures: A Pseudo-code Approach with C, Gilberg & Forouzan, Thomson Learning
5. Data Structures using C & C++, Ten Baum, Prentice Hall International

ONLINE RESOURCES:

1. <https://nptel.ac.in/courses/106102064>
2. https://onlinecourses.swayam2.ac.in/cec19_cs04/preview

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25ES2EE104) ELEMENTS OF ELECTRICAL AND ELECTRONICS ENGINEERING LABORATORY

TEACHING SCHEME		
L	T/P	C
0	2	1

EVALUATION SCHEME					
D-D	PE	LR	CP	SEE	TOTAL
10	10	10	10	60	100

COURSE PRE-REQUISITES: Mathematics, Physics

COURSE OBJECTIVES:

- To recognize different circuit reduction techniques
- To understand the construction of electrical equipment and operation
- To practice the techniques to control and assess electrical machines and installations
- To learn principles of operation, construction and characteristics of various electronic devices

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Apply different network reduction techniques to solve and analyze electrical circuits

CO-2: Realize the compatibility of electrical machines in different engineering fields & Installations

CO-3: Control different electrical machines and evaluate their performance

CO-4: Apply basic electronics device in real time applications

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-2	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-3	2	2	1	-	-	-	-	1	-	-	-	2	1
CO-4	2	2	1	-	-	-	-	1	-	-	-	2	1

LIST OF EXPERIMENTS:

1. Demonstration of safety precautions, measuring instruments, Multimeters, Function Generator, RPS, CRO
2. Demonstration of LT switchgear components
3. Verification of KVL & KCL
4. Verification of the superposition theorem
5. Analysis of series RL, RC, and RLC circuits
6. Control of voltage of an alternator by its excitation
7. Load test on 1- ϕ transformer

8. Brake test on 3- ϕ induction motor
9. Speed control of Dc shunt motor
10. Diode Characteristics
11. Transistor CE Characteristics (Input and Output)
12. Half Wave & Full Wave Rectifiers without filter

TEXT BOOKS:

1. Basic Electrical Engineering, D. C. Kulshreshtha, 2nd Edition, TMH, 2019
2. Basic Electrical Engineering, P. Ramana, M. Suryakalavathi, G. T. Chandra Sekhar, 1st Edition, S. Chand, 2018
3. Electronic Devices and Circuits, S. Salivahanan and N. Suresh Kumar, 3rd Edition TMH, 2019

REFERENCES:

1. Electrical and Electronics Technology, E. Hughes, 10th Edition, Pearson, 2010
2. Electrical Engineering Fundamentals, Vincent Deltoro, 2nd Edition, Prentice Hall India, 1989
3. Electrical and Electronics Measurements and Instrumentation, A. K. Sawhney, 3rd Edition, Dhanpat Rai & Co., 1983
4. Basic Electrical Engineering, D. P. Kothari and I. J. Nagrath, 3rd Edition, Tata McGraw-Hill, 2010
5. Engineering Circuit Analysis, William Hayt and Jack E. Kemmerly, 8th Edition, McGraw-Hill, 2013

ONLINE RESOURCE:

1. <https://nptel.ac.in/courses/117106108>

VNR VIGNANA JYOTHI INSTITUTE OF ENGINEERING AND TECHNOLOGY

B.Tech. II Semester

(25MN6HS102) ENVIRONMENTAL SCIENCE

TEACHING SCHEME		
L	T/P	C
2	0	0

EVALUATION SCHEME		
SE-I	SE-II	TOTAL
50	50	100

COURSE PRE-REQUISITES: Basic Knowledge on Environmental Issues

COURSE OBJECTIVES:

- To recognize the principles of environment
- To identify the sources and causes of environmental pollution
- To analyze the causes of climate change and identify climate change mitigation and adaptation technologies
- To list out the benefits in creating circular economy and sustainability
- To emphasize the environmental conventions, policies and regulations

COURSE OUTCOMES: After completion of the course, the student should be able to

CO-1: Gain knowledge about principles of Environment and allied problems

CO-2: Identify the causes and suggest remedial measures for Environmental Pollution

CO-3: Appraise the mitigation measures and adaptation technologies related to climate change

CO-4: Familiarize with the importance of circular economy towards sustainability

CO-5: Relate and implement environmental policies and regulations

COURSE ARTICULATION MATRIX:

(Correlation of Course Outcomes with Program Outcomes and Program Specific Outcomes using mapping levels 1 = Slight, 2 = Moderate and 3 = Substantial)

CO	PROGRAM OUTCOMES (PO)											PROGRAM SPECIFIC OUTCOMES (PSO)	
	PO-1	PO-2	PO-3	PO-4	PO-5	PO-6	PO-7	PO-8	PO-9	PO-10	PO-11	PSO-1	PSO-2
CO-1	2	2	1	-	1	2	2	1	1	1	2	-	-
CO-2	2	2	1	-	1	2	2	1	1	1	2	-	-
CO-3	2	2	1	-	1	2	2	1	1	1	2	-	-
CO-4	2	2	1	-	1	2	2	1	1	1	2	-	-
CO-5	-	1	1	-	-	2	1	1	1	1	-	-	-

MODULE 1:

Introduction to Environmental Science: Importance and Scope of Environmental Science. Overview of Environment & its components. Ecosystem function and uses. Biodiversity – Types and Values.

MODULE 2:

Synergy With Environment: Relationship between environmental science & society – Influence of Industry & infrastructure on environment (Specific case studies). Sources, Impacts and prevention of air, water and soil pollution.

MODULE 3:

Climate Change: Science behind climate change – factors responsible for climate change, Role of greenhouse gases – Global temperature rise & its impact on environment & human health. Carbon footprint – Briefing on Paris agreement, Identify key sectors for low carbon footprint. Climate change mitigation & adaptation strategies.

MODULE 4:

Moving Towards Sustainability: EIA and EMP - importance, Sustainable agriculture – Importance of Organic farming and hydroponics. Role of AI & IOT for efficient management of environmental issues – Health, air, water, and soil. Sustainable living practices – Principles and methods (Linear Economy vs circular economy). Green Buildings.

MODULE 5:

Policies and Regulations: International conventions: Montreal Protocol, COP15, Kyoto protocol (Salient Features). National Environmental Policies – Salient features of EPA act, Air act, Water Act, Forest and Wildlife act. National Green Tribunal Act (2010)- Features and Significance. Extended producer responsibility. Innovative approaches to waste management: Plastic, tyre and batteries.

TEXT BOOKS:

1. Living in the Environment, G. Tyler Miller, Scott E. Spoolman, 20th Edition, Cengage Learning
2. Environmental Studies, Rajagopalan, Oxford University Press, 2005
3. Introduction to Climate Change, Andreas Schmittner, Oregon State University, 2018 <https://open.oregonstate.edu/climatechange/>

REFERENCES:

1. Green Development: Environment and Sustainability in a Developing World, Bill Adams, 4th Edition, Routledge Publishers, 2021
2. Fixing Climate, Robert Kunzig & Wallace S. Broecker, Profile Books Publisher, 2009
3. Plastic Waste and Recycling-Environmental Impact, Societal Issues, Prevention and Solutions, 1st Edition, edited by Trevor Letcher, Academic Press, 2020
4. Waste Management Practices: Municipal, Hazardous, and Industrial, John Pichtel, CRC Press, 2020
5. Environmental Science: A Global Concern, William P. Cunningham, Mary Ann Cunningham, McGraw-Hill Education, 2022

ONLINE RESOURCE:

1. https://www.google.co.in/books/edition/Waste_Management_Practices/pG3SBQAAQBAJ?hl=en&gbpv=1&pg=PA1&printsec=frontcover